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Roll No

EC-8001 (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
VLSI Design
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions out of eight questions.
ii) All questions carry equal marks.
iii) Assume suitable data, if required.

1. a) Give a brief introduction on IC. Write and explain the classification of different IC's.
b) Explain Microelectronics Field. Explain the complexity of integrated circuits.
2. a) How you can explain the MOS fabrication using various steps? Explain any two of them.
b) Write down some of the differences between the Bipolar technology and Hybrid Technology.
3. a) Write and explain the different design rules in MOS technology.
b) How you can differentiate the designing parameters between PMOS and NMOS processing steps? Explain with suitable example.
4. a) Define device modeling. Discuss the DC models and its classification.
b) Draw and explain High frequency small signal in MOS using any of the DC circuit.
5. a) How you can explain the simulation using SPICE with suitable flowchart?
b) Discuss about the Structured Logic Forms. How is it different from Random Logic Forms?
6. a) Explain the coding process in Micro coded controllers using any of the example.
b) What is the Bit Serial Processing Elements? Explain its circuit or architecture with detail.
7. a) Draw and Explain Twin Tub process in detail.
b) Explain Latch up triggering. Latch up prevention and Internal Latch up prevention techniques.
8. Write short notes (any four):
 - a) Interconnects
 - b) Quasi Static Register Cells
 - c) Algotronix
 - d) Basic and well CMOS process
 - e) Epitaxy
